Installing SUSE Linux Enterprise Server Manually Using Agama

WHAT?

This article describes how to install SUSE Linux Enterprise Server using Agama. The information presented here applies to installation of the products on both bare metal and virtual machines.

WHY?

Read this article to understand the process of installing SUSE Linux Enterprise Server using Agama.

EFFORT

You may need 15 minutes to read this article.

GOAL

Learn how to manually install SUSE Linux Enterprise Server using Agama.

REQUIREMENTS

- A bare metal server or a virtual machine. For server installations without any desktop environment, SUSE recommends a minimum of 1 CPU, 2 GB memory and 32 GB storage (which includes storage for Btrfs snapshots in the root partition, swap space, and storage for software packages).
- An active registration code for the product you want to install. You can generate a registration code for the product and activate its subscription for your organization at the SUSE Customer Center.

Note: Optional registration

Certain images that are signed with the developer's key may allow you to skip registration before or while installation. Besides, certain images may contain all installable packages for your operating system that you can use as an offline package repository. In such cases, you may not need an active registration code before installation. However, if you use software packages from the official online repositories, SUSE recommends registering your product with the SUSE Customer Center.

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1 Introduction to Agama

This section provides a short overview of Agama, which is the default installer for SUSE Linux Enterprise Server 16.0 and later versions.

1.1 What is Agama?

Agama is a flexible and powerful open source installer developed by SUSE. You can use it to perform manual or automated installation of supported SUSE Linux Enterprise Server versions on a variety of machine architectures. The Agama-based installation method provides an intuitive interface for *ready-to-use* images for different use cases and workloads. In combination with Cockpit and Ansible, it substitutes YaST in SUSE Linux Enterprise Server.

2 Installing SUSE Linux Enterprise Server using Agama

This section describes how to install SUSE Linux Enterprise Server using the Agama-based ISO image file.

2.1 Introduction

The following sections describe the Agama-based manual installation procedure for SUSE Linux Enterprise Server. The information applies to installation on both bare metal and virtual machines.

2.2 Requirements

- An Agama-based ISO image file for supported versions of SUSE Linux Enterprise Server. You can download it from the SUSE Customer Center and verify its checksum to ensure its integrity.

Note: Available images for products

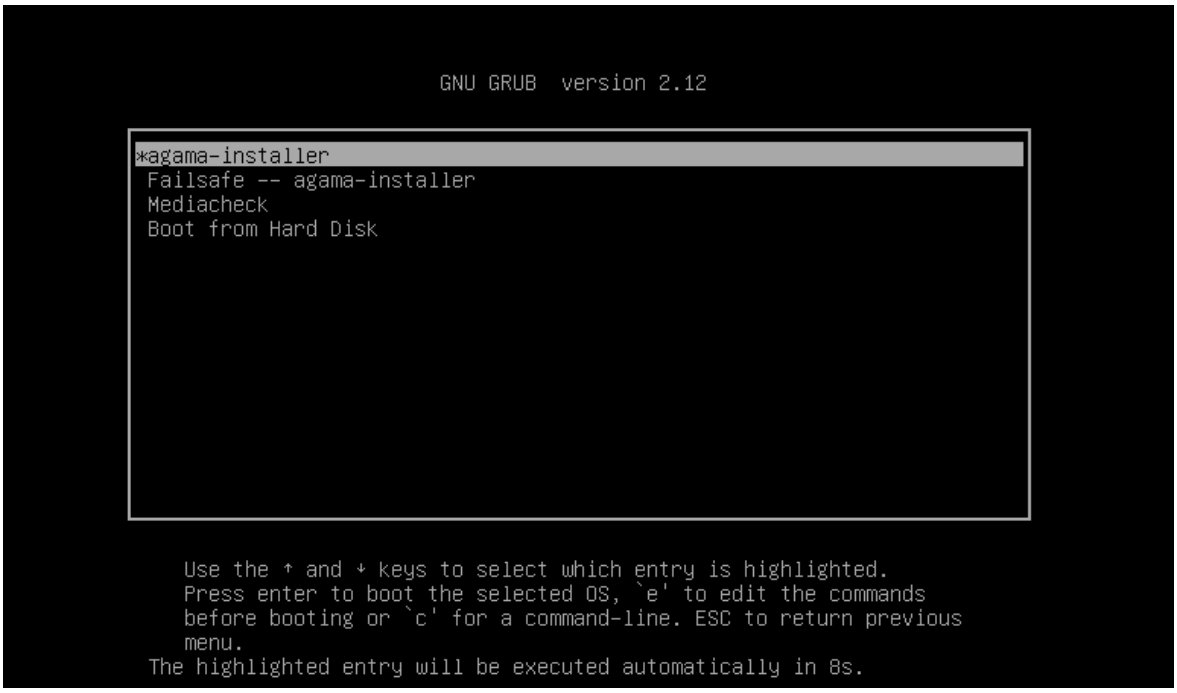
The Agama-based installation image is available for SUSE Linux Enterprise Server 16.0 and later versions.

2.3 Manually installing SUSE Linux Enterprise Server for x86_64 using Agama

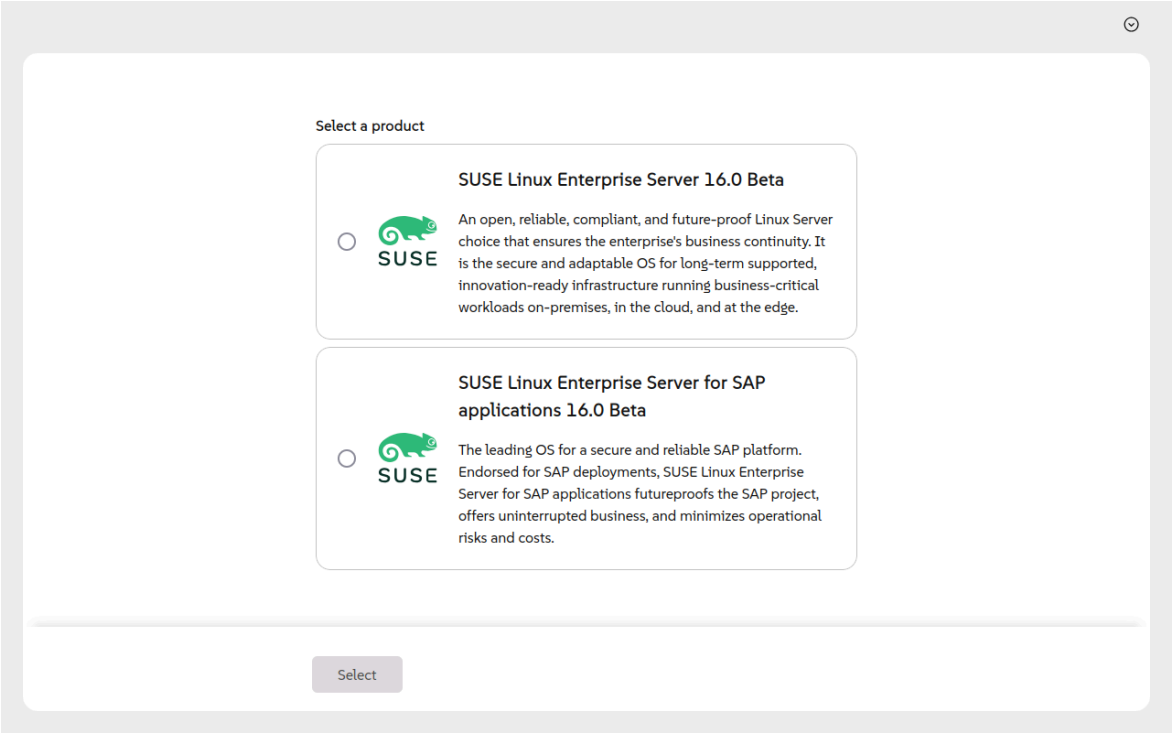
To manually install SUSE Linux Enterprise Server using Agama, perform the following steps:

PROCEDURE 1: MANUAL INSTALLATION OF SUSE LINUX ENTERPRISE USING AGAMA

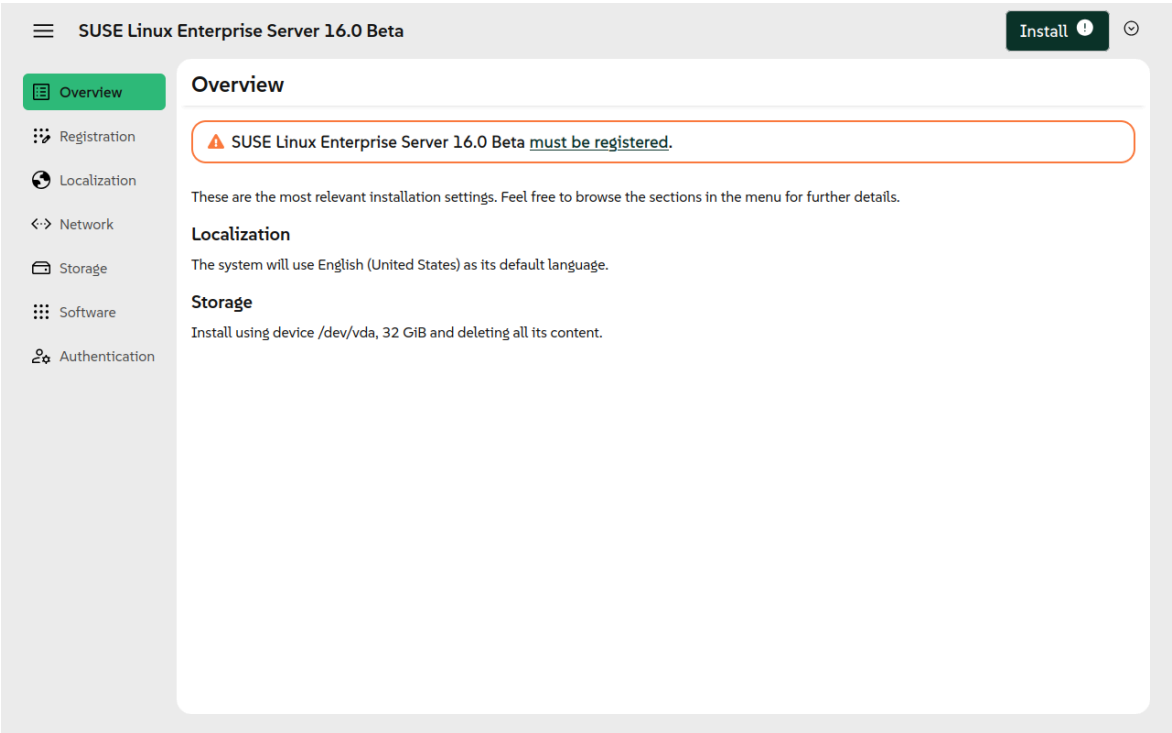
1. Select a bootable media and start the installation process.
   - For a bare metal server, select the device or storage volume from which to boot.
   - For a virtual machine guest, select the path to the image file and create a virtual machine with a configuration that is appropriate for the anticipated workload.
2. In the GRUB 2 boot menu, select *agama-installer*.



3. Select SUSE Linux Enterprise Server 16.0 for installation and accept the license to continue.



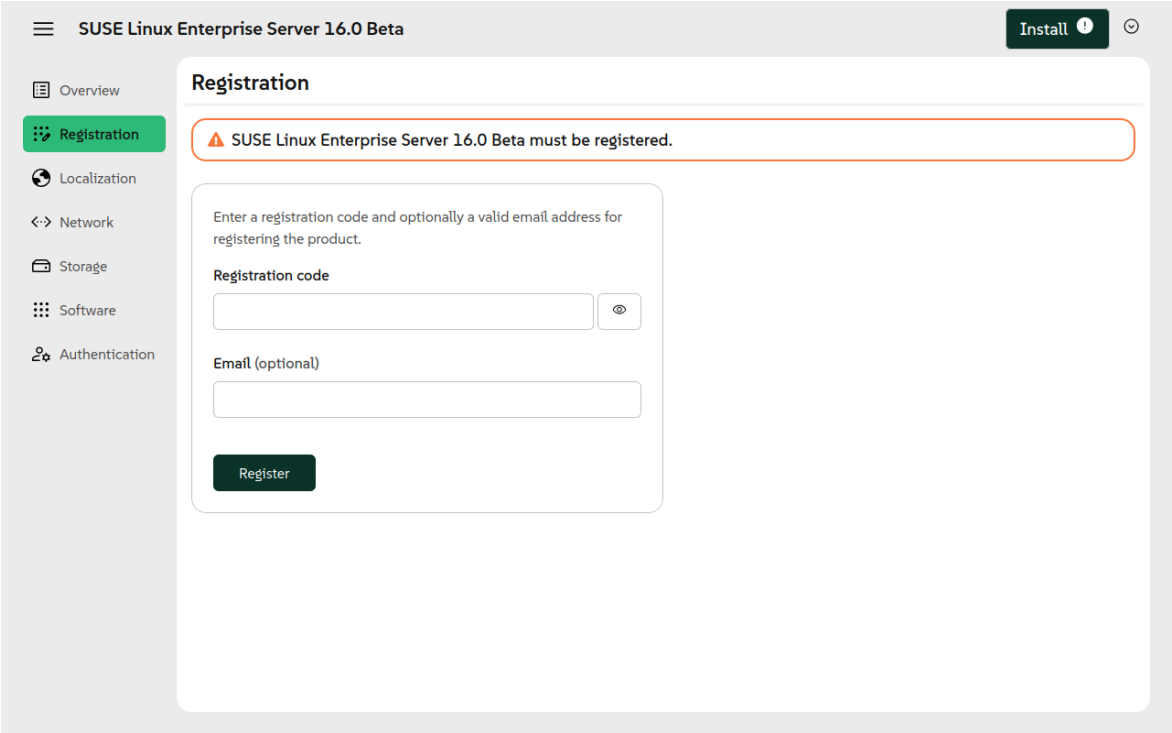
4. The overview with the most relevant installation settings is shown. Adjust the settings by selecting the category from the menu on the left or directly proceed with the installation.



Tip: Installer language and keyboard

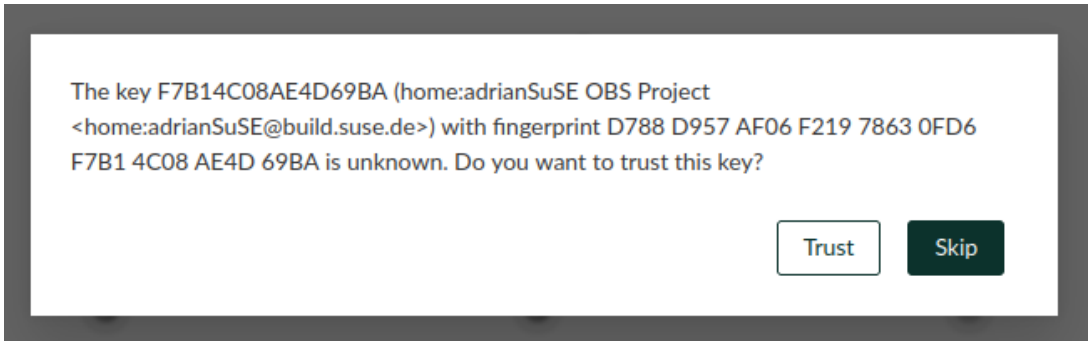
Click the arrow menu in the top right corner and select *Installer options* to change the language and keyboard layout used by the installer. The language can only be changed on local installations. If you are using a web browser for a remote installation, configure the preferred content language in the browser settings and reload the installer.

5. Register your product. Enter a registration code and optionally a valid email address.



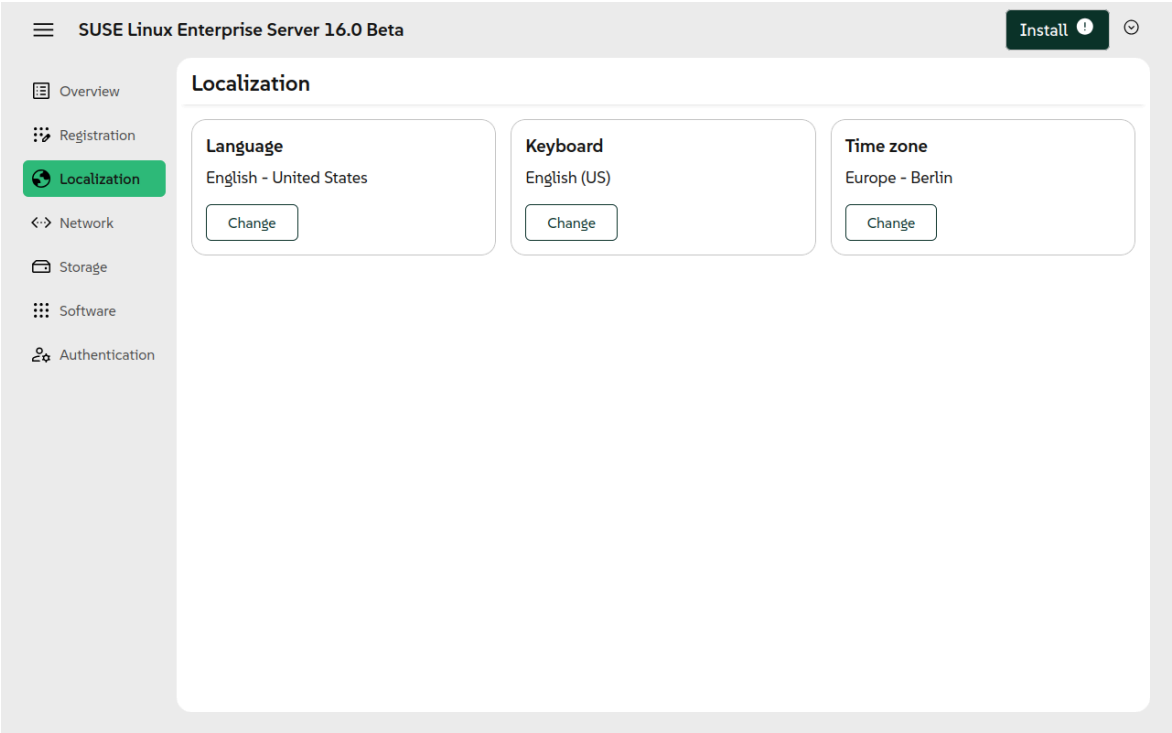
Note: Optional registration for images signed with the developer's key

Images signed with the developer's key may allow you to skip registration before installation. If your product's image supports installation without registration, you may see a pop-up similar to the following:

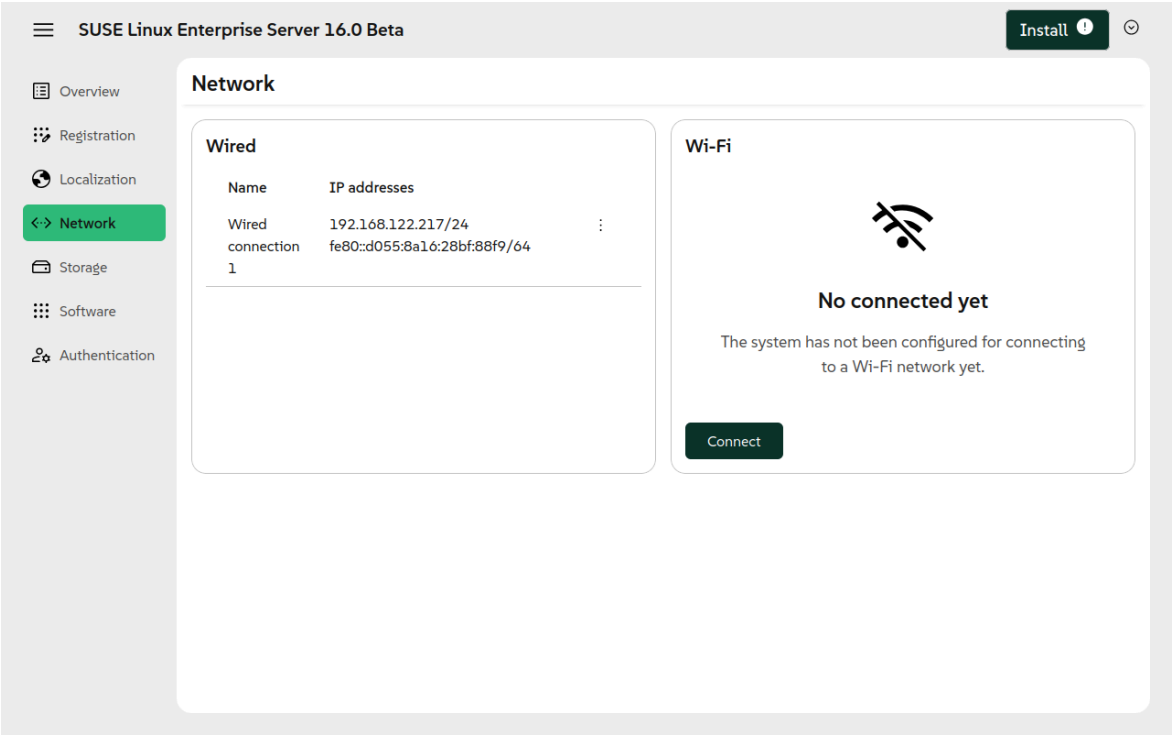


To install without registration, select *Trust* on the pop-up.

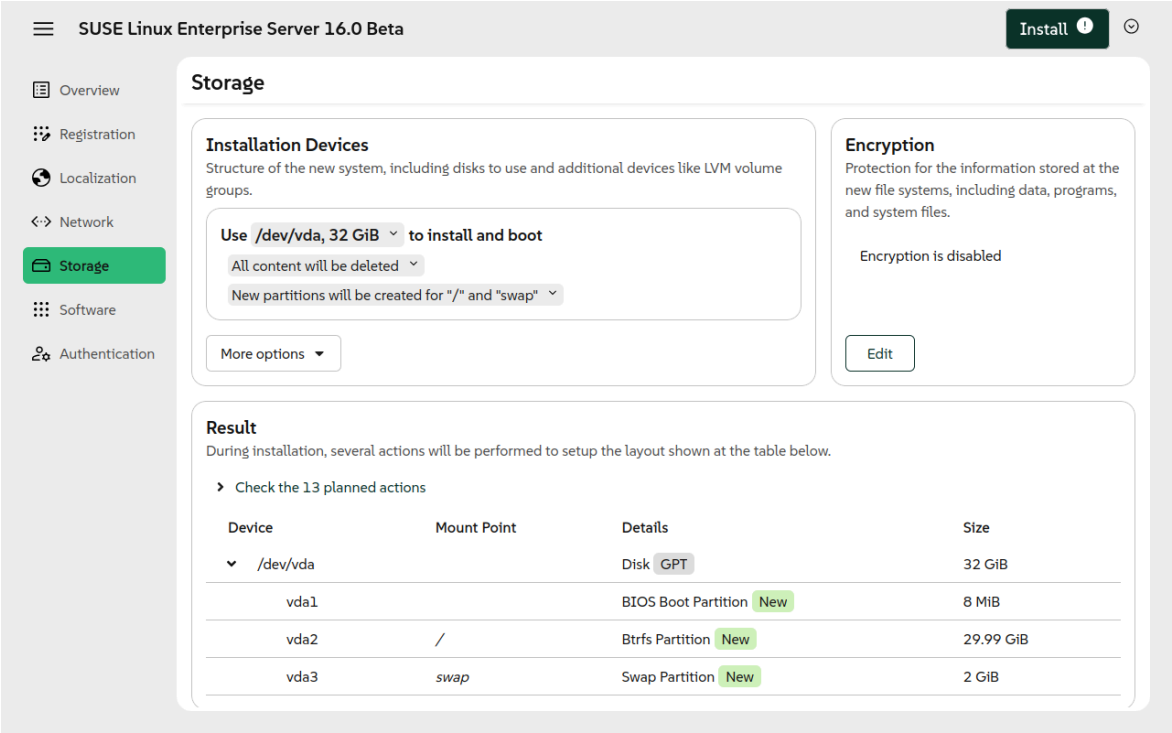
6. *(Optional)* Adjust your localization settings such as language, keyboard layout and time zone.



7. *(Optional)* Configure your network settings. To edit an existing network connection, click the three dots.



8. Select a storage device for installation.



Currently installations on single discs or LVM are supported. To change the disk or use LVM, click the device name *More options* button in the *Installation devices* section.
To detect an iSCSI disk, select the iSCSI initiator and click *Discover iSCSI tagets*.
By default, the file systems will be allocated as new partitions on the selected device. You can change this behavior in the drop-down list labeled *All content will be deleted* drop-down list in the *Installatin devices* section. The following space policies are available:

SPACE POLICIES

Delete current content

All partitions will be removed and any data on the disks will be lost.

Shrink existing partitions

The data is kept, but current partitions will be shrinked as needed.

Use available space

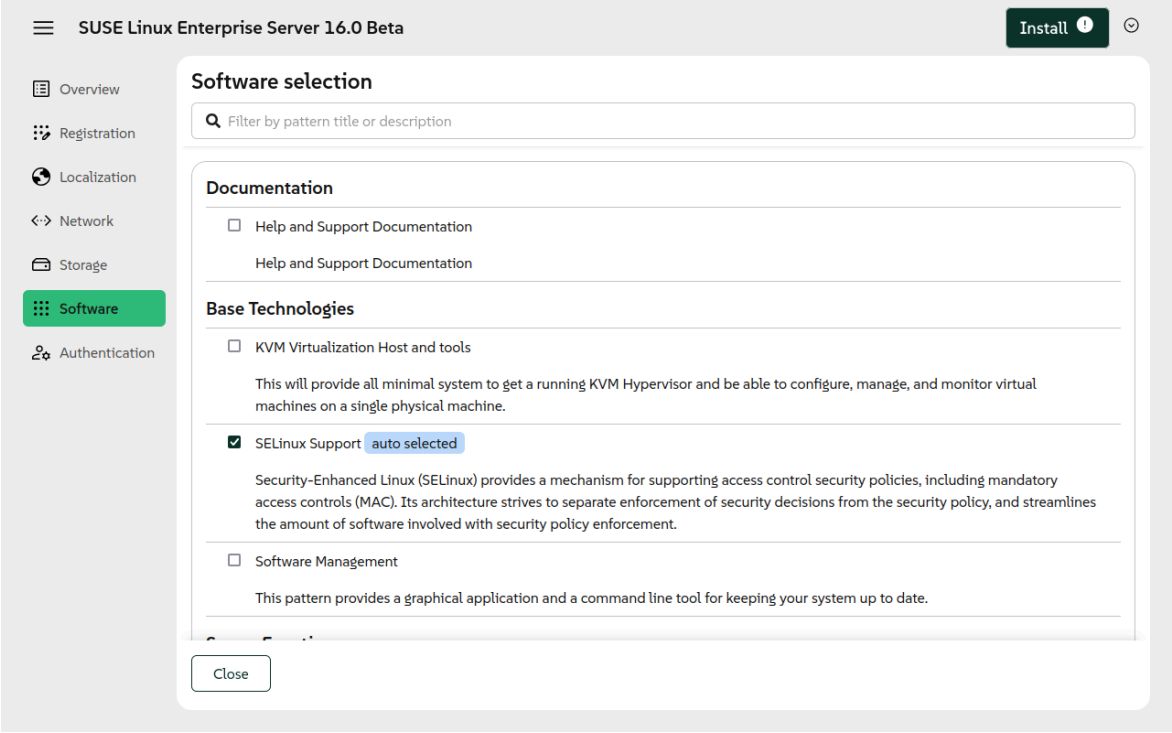
The data is kept. Only the space not assigned to any partition will be used.

Custom

Select what to do with each partition.

To edit individual partitions, expand the *Partitions and file systems* section by clicking on the > Optionally, you can configure Btrfs snapshots, boot options and enable full disk encryption (FDE).

9. *(Optional)* Add additional software to install such as the KVM hypervisor or Cockpit system management service.

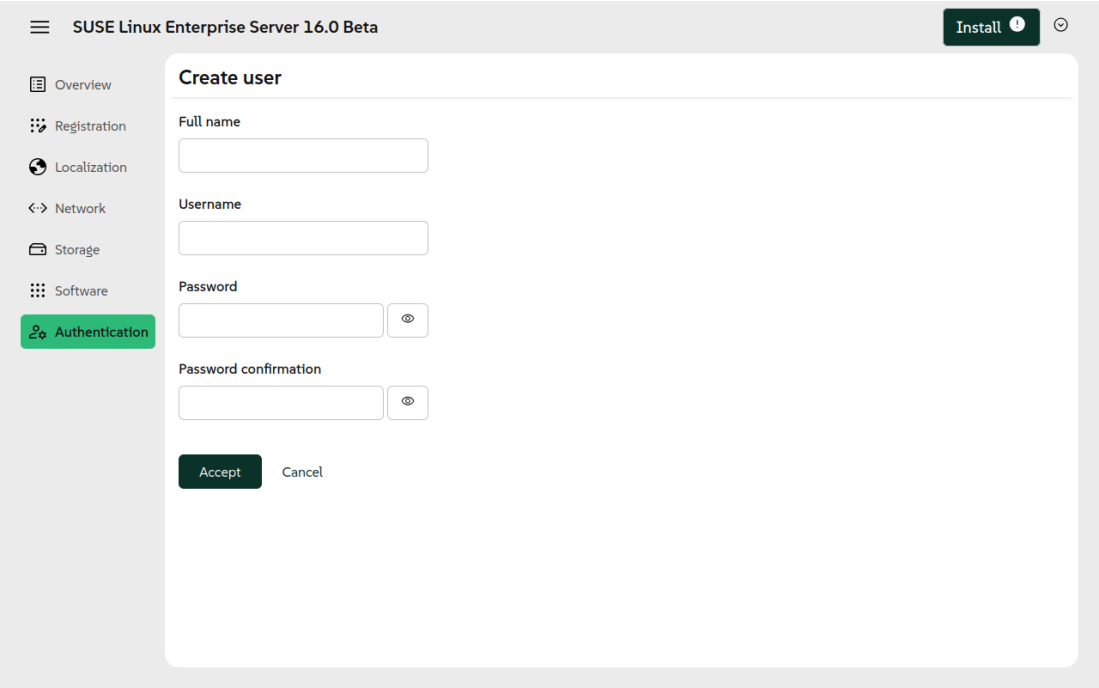


10. Configure the authentication settings.

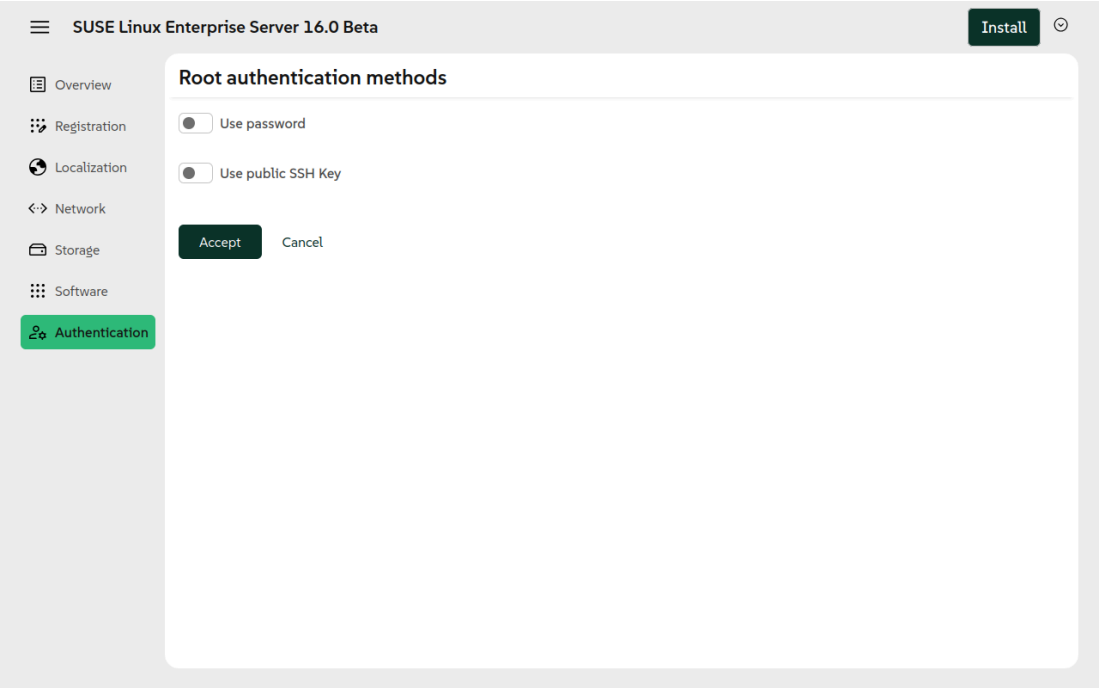
Important: No SSH `root` access

SSH access for `root` is disabled by default. Therefor it is strongly recommended to create a system user. The first system user is granted **`sudo`** privileges automatically and can also be used for accessing Cockpit.

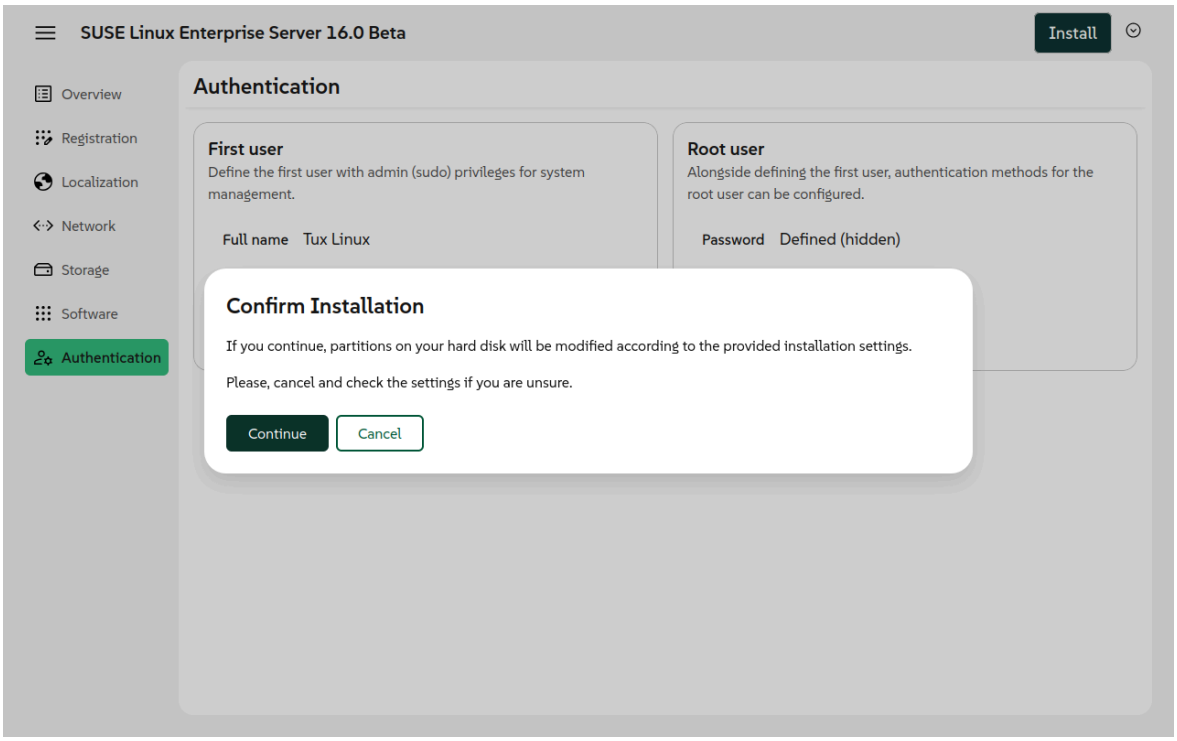
a. Create a non-root user. Click *Define a user now* and enter details such as user name and password.



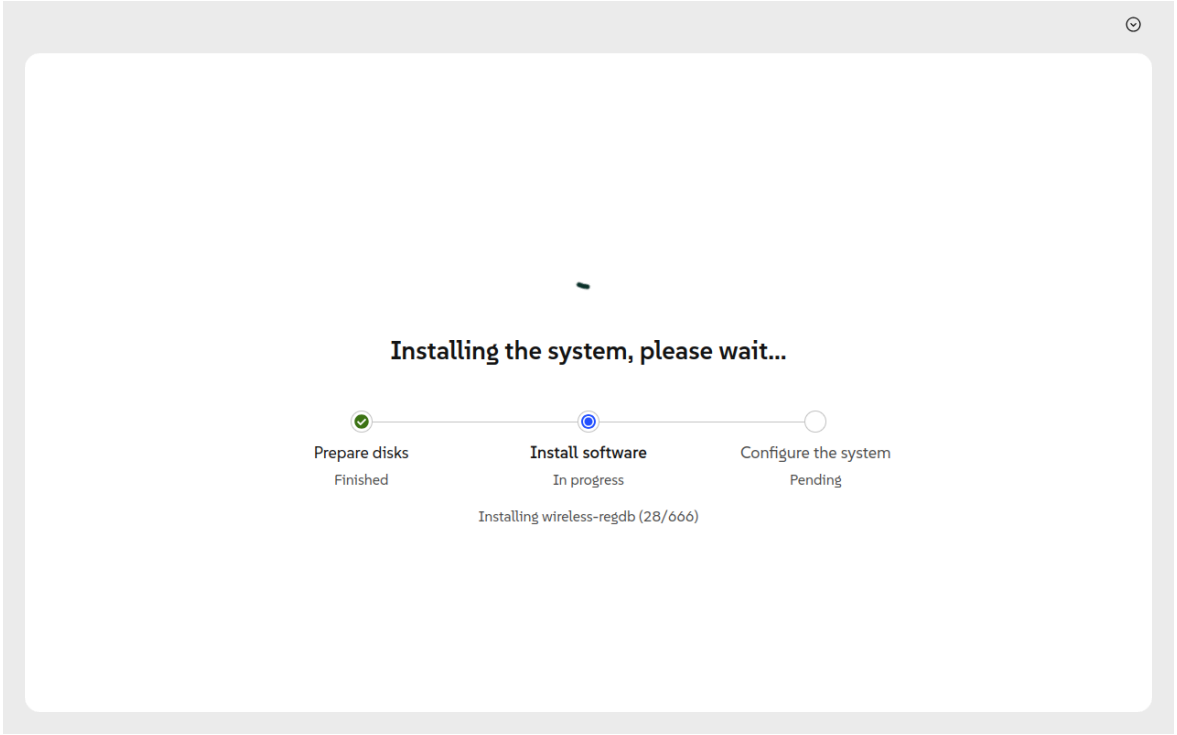
b. Configure the authentication methods for the `root`. You can use a secure password, a SSH public key, or both.



11. Click *Install* and confirm to start the installation.



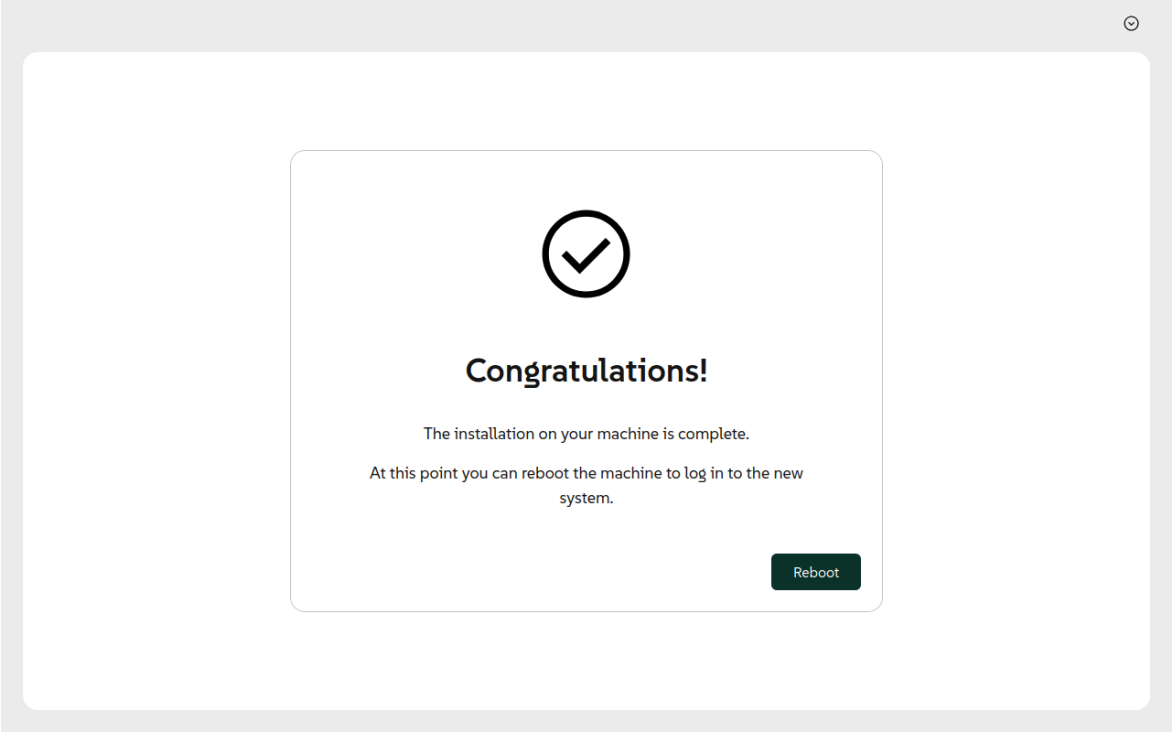
12. Wait for the system installation to complete. It takes around 30 minutes to complete the installation.



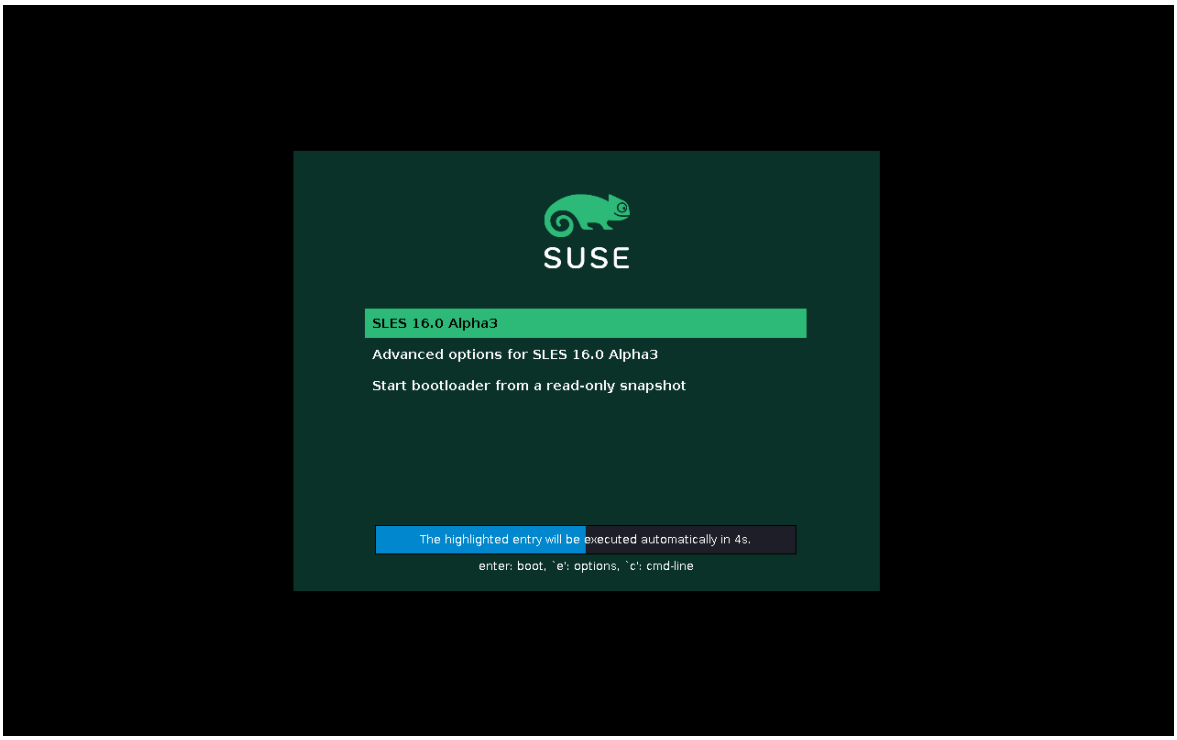
Tip: Installation logs

You can use the arrow menu in the top right corner of the window to view the logs and system messages.

13. Once the installation is complete, select *Reboot*.



14. Using the GRUB 2 boot menu, boot into the newly installed system.

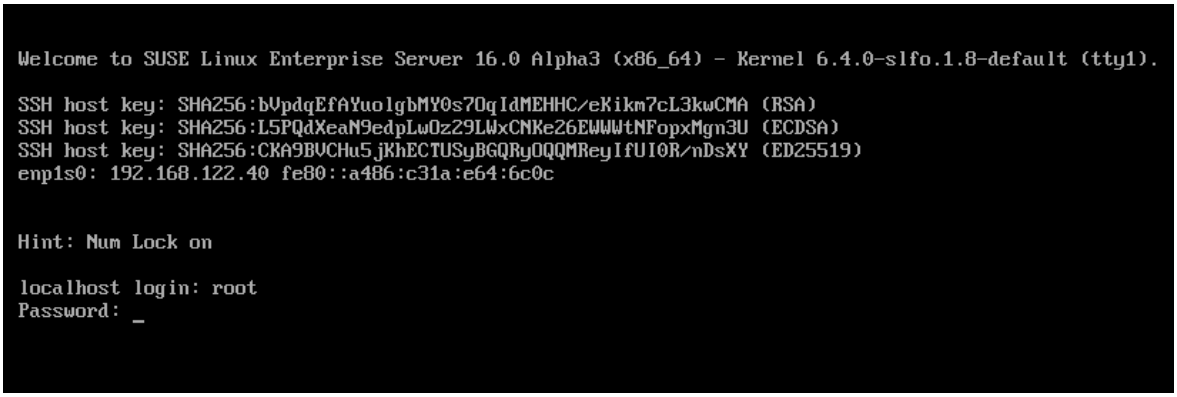


2.4 Summary

After successful configuration checks, product registration, installation and reboot, verify that you can log in and use your system.

PROCEDURE 2: VERIFYING THE SYSTEM AFTER INSTALLATION

1. Log in to the system as root.



2. To verify that your system is working as expected, run a few basic commands.

```
Welcome to SUSE Linux Enterprise Server 16.0 Alpha3 (x86_64) - Kernel 6.4.0-slfo.1.8-default (tty1).

SSH host key: SHA256:bVpdqEfAYuolgbMY0s7OqIdMEHHC/eKikm7cL3kwCMA (RSA)
SSH host key: SHA256:L5PQdXeaN9edpLwOzZ9LWxCNKe26EWWWtNFopxMgn3U (ECDSA)
SSH host key: SHA256:CKA9BVCHu5jKhECTUSyBGQRyOQQMReyIfUI0R/nDsXY (ED25519)
enp1s0: 192.168.122.40 fe80::a486:c31a:e64:6c0c


Hint: Num Lock on

localhost login: root
Password:
Have a lot of fun...
localhost:~ # whoami
root
localhost:~ # pwd
/root
localhost:~ #
```

3 For more information

For information on related topics, refer to the following resources:

- Documentation of the Agama open-source project: https://agama-project.github.io/docs

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